

CodioProjectFileEditFindViewToolsEducationHelpConfigure...Project Index (static)Configure...

imgtxt.pyx1import os2import openai3from PIL import Image, ImageOps4from io import BytesIO5import requests6import secret78# Set API key and prompt9openai.api_key = secret.key1011# WRITE YOUR CODE HERE

100% (1:1)

Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage

* Canonical Livepatch is available for installation.
- Reduce system reboots and improve kernel security. Activate at:
https://ubuntu.com/livepatch
*
* Welcome to the Codio Terminal!
*
* https://docs.codio.com/develop/develop/ide/boxes/overview
*
* Your Codio Box domain is: yankeetrick-avenuenumeric.codio.io
*
Last login: Fri Jul 21 21:21:11 2023 from 192.168.11.179
codio@yankeetrick-avenuenumeric:~/workspace\$

GuidexCollapse

Image-to-Text Application

For an image-to-text application, you can use DALL-E 2 to generate a variety of images and then use GPT-3 to describe the images or provide captions.

Use DALL-E 2 API to generate an image based on a text prompt:

```
# Generate the base image
response = openai.Image.create(
    prompt="cute kitten playing with a ball of yarn",
    n=1,
    size="512x512"
)

image_url = response['data'][0]['url']
print(image_url)
```

TRY IT!

Command was successfully executed.

Feel free to copy and paste the image URL to compare.
Use GPT-3 to generate a description or caption for the generated image:

```
# Generate a description with GPT-3
response = openai.Completion.create(
    engine="text-davinci-002",
    prompt=f"Describe the following image: [img]{image_url}[/img]",
    max_tokens=50,
    n=1,
    stop=None,
    temperature=0.7,
)

image_description = response.choices[0].text.strip()
print(image_description)
```

TRY IT!

Click here to refresh your image

It's important to be aware that image-to-text applications may not always provide the most accurate descriptions or captions. There are several reasons for this:

Lack of visual context: GPT-3 is a text-based model and does not have direct access to the actual image content. Instead, it relies on the image URL, which may not always provide enough context for the model to generate a precise description.

Ambiguity in images: Images can often contain ambiguous or abstract elements, making it difficult for the model to understand the exact content or context of the image. This can result in inaccuracies or vague descriptions.

Model limitations: While GPT-3 is an advanced language model, it is not perfect and has its limitations. It may not always be able to generate the most accurate or relevant descriptions, especially for images with complex or highly specialized content.

To improve the accuracy of image-to-text applications, you can consider using specialized computer vision models, such as **OpenAI's CLIP**, which is designed specifically for tasks involving both text and images. These models are better equipped to handle the challenges associated with understanding and describing visual content.

GPT-3 & image

(True or False):
GPT-3 has direct access to the actual image content when generating a description or caption for an image.

☐ True

☒ False

GPT-3 is a text-based model and does not have direct access to the actual image content. Instead, it relies on the image URL, which may not always provide enough context for the model to generate a precise description.

Check It!

Next